

Pneumonia Detection Using Deep Learning and Machine Learning Approaches from Chest X-Ray

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Abstract—Pneumonia is a lung disease that causes an infection that inflames the air sacs (alveoli) in one or both lungs. It is a very common problem in both children and the elderly. It can create a serious health issue, even death, if not detected in time. This study utilizes deep learning and Machine Learning techniques to detect pneumonia accurately. The proposed model identifies pneumonia accurately at the first stage. CNN-based Deep Learning models like DenseNet121, DenseNet201, EfficientNetB0, and ResNet152V2 are used as a feature extractor. Additionally, A custom Deep Learning-based classifier, and Machine Learning classifiers including XGBoost, and Support Vector Machine (SVM) are utilized. Various pre-processing techniques, such as Contrast Limited Adaptive Histogram Equalization (CLAHE), Laplacian filtering, etc., on chest X-ray images from two publicly available datasets have been applied to enhance the detection. The best performance was achieved using DenseNet201 combined with the Deep Learning-based Custom classifier, achieving an accuracy of 99.68% on dataset-1. Correspondingly, the combination of ResNet152V2 and the Deep Learning-based custom classifier achieved an accuracy of 98.75% on dataset-2. This study demonstrates the effectiveness of Deep Learning techniques in pneumonia detection.

Index Terms—Pneumonia Detection, Classification, Deep Learning, Machine learning, Digital Image processing.

I. INTRODUCTION

Pneumonia is a lung disease that creates difficulty breathing. Children and the elderly can not survive in this situation for a long time. Fast and accurate detection can help to save their life. Manually detecting Pneumonia is a time-consuming problem, more difficult to find the problem accurately, and can cause human error. In 2021, pneumonia claimed the lives of 2.2 million, including 502,000 children, according to the Global Burden of Disease.[1]. From Figure 2, we can see that in 2022, the total number of deaths from influenza and pneumonia in Canada amounted to 5,985, an increase from 4,115 deaths the year before. This figure presents a subset of data taken from a dataset from 2000 to 2022 [15]. Automated detection can solve this gap by providing fast and accurate reports.

In recent years, the Deep learning technique has shown the best performance in the medical image classification task, including pneumonia detection. Pre-trained models have the

most powerful feature extractive capability, which can learn complex patterns from raw images. Chest X-ray images are used for Pneumonia detection. X-ray tests are cost-effective compared to other tests and can be efficiently used in image processing. Pre-processing methods (CLAHE and Laplacian filtering) can be used to increase image quality. Those methods help to identify the best solution.

Pre-trained CNN models like DenseNet121, DenseNet201, DenseNet201, EfficientNetB0, and ResNet152V2 can be used to efficiently extract features. As well as here, SVM, XGBoost, and Custom Deep Learning Classifiers can be used for model training and help to evaluate Pneumonia detection accurately from chest x-ray images.

This paper is organized as follows: Section II provides related work in pneumonia detection using deep learning and machine learning. Section III tells the methodology, including data pre-processing, feature extraction, and classification. Section IV provides the Output of this paper. Section V contains the limitation & future work part, and Section VI provides the conclusion.

II. RELATED WORK

S. Sharma et al have used A Deep Learning model for the Detection of Pneumonia from Chest X-ray images using VGG-16 and Neural Networks. The dataset contains 5,863 images of chest X-rays, which have been divided into two classes: pneumonia and normal. They got acceptable accuracy. [3]

M. S. A. Reshan et al. have used the Detection of pneumonia from chest X-ray images utilizing the MobileNet model. They used ResNet50, ResNet152V2, DenseNet121, DenseNet201, Xception, VGG16, EfficientNet, and MobileNet models. They used two datasets, 5856 images and 112,120 images of chest X-rays.[4]

F. Wahid and S. Azhar used the Detection of pneumonia from chest X-ray images utilizing the MobileNet model. They used KNN, SVM, DT, sRBM, and eRBM models. They have used three datasets, having 1500 images, 3000 images, and 1248 images of chest X-rays. [5]

N. C. Kundur used Detection of Pneumonia in chest X-rays using transfer learning. They used VGG16, ResNet152V2,

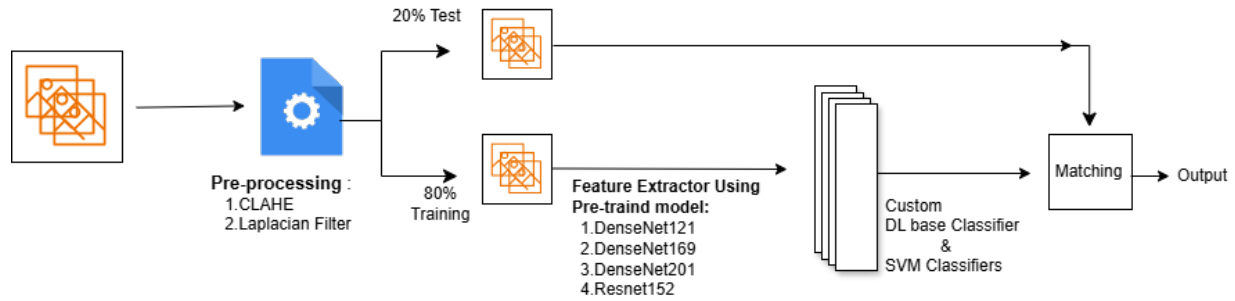


Fig. 1: Methodology Diagram

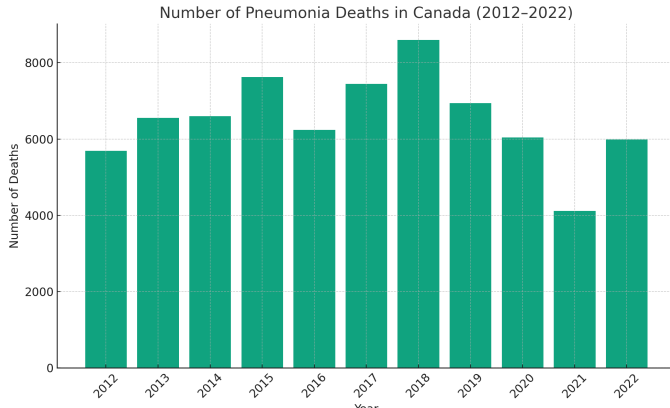


Fig. 2: Total number of deaths from pneumonia in Canada [15]

DenseNet, ResNet-50, InceptionNet, and Custom CNN models. [6]

Kailasam, R., and Alasubramanian, S. worked on Deep Learning for Pneumonia Detection: A Combined CNN and YOLO Approach using chest x-ray images. This work focuses on distinguishing between normal and pneumonia. The YOLO algorithm helps to identify the region of interest. They used a publicly available dataset that contains a total of 5863 chest x-ray images and got good output.[7]

Rabbah and Ridouani worked on improving pneumonia diagnosis with high-accuracy CNN-based chest X-ray image classification and integrated gradient. They were focused on achieving high accuracy and used 5,856 images as input data. They utilized the inception_v3 layer for feature extraction, combined with dense layers to build a high-performing binary classification.[8]

Yas, M., and Mehta considered a 'Deep Learning for Pneumonia Diagnosis: A Custom CNN Approach with Superior Performance on Chest Radiographs' system. This study proposes a pneumonia detection method to automate the process using deep learning techniques. They used those (VGG19, ResNet50, InceptionV3, DenseNet201, and MobileNetV3) models to achieve high accuracy. Those models worked on 5863 X-ray images.[9]

Billah, Muhammad, and Al Rakib worked on 'A Comparative Study on the Detection of Pneumonia in Chest X-Ray

Images Utilizing Deep Learning Models'. Dataset 5,863 chest X-ray images collected from Kaggle. Those (VGG16, Inception V3, and ResNet 50V2) models were used for achieving high accuracy.[10]

Angara, S., and Mannuru worked on a Pneumonia prediction model, which is considered A novel method to enhance pneumonia detection via a model-level ensembling of CNN and vision transformer. They used 1,341 normal and 3,875 pneumonia chest x-ray images for the training model. ResNet34 variant and a Multi-Axis Vision Transformer are considered as models to achieve high performance.[11]

Singh, S., and Kumar, M proposed Efficient pneumonia detection using Vision Transformers on chest X-rays. 5863 X-ray images and the Vision Transformer (ViT) architecture are used for efficient results.[12]

Mabrouk, A., and Redondo proposed Pneumonia Detection on chest X-ray images using an Ensemble of Deep Convolutional Neural Networks. CNN pre-trained (DenseNet169, MobileNetV2, and Vision Transformer) models used 5,856 chest x-ray images for training.[13]

III. METHODOLOGY

This section shows the working process or methodology of the research step by step.

A. Methodology diagrams

Figure 1 shows the overall procedure. First, the need to collect data. Then we have used a combination of pre-processing techniques. Then we used 80% images for model training purposes, and 20% images were reserved for model testing or evaluating model accuracy. After pre-processing, a feature extractor is used to extract features from images. Then, classifiers are used for binary classification.

B. Description

1) *Data Collection:* In this paper, we have used two publicly available datasets.

The first dataset we used for this study contains a total of 5,216 chest X-ray images, categorized as normal (1,341 images) and pneumonia-affected (3,875 images)[2]. The dataset was obtained from Kaggle and preprocessed to ensure consistency and quality. The other Chest X-ray dataset contains 8000 images. 4000 images are equally distributed between

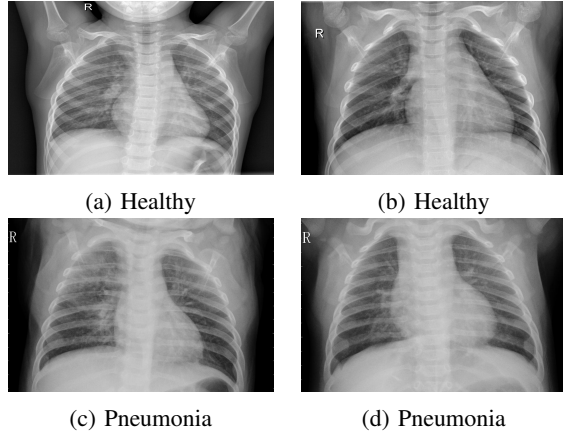


Fig. 3: Sample X-ray images from Dataset 1

the healthy and pneumonia classes.[14] The table I below summarizes the dataset distribution:

TABLE I: Dataset distribution of chest X-ray images.

Category	Dataset 1	Dataset 2	Total
Normal	1,341	4,000	5,341
Pneumonia	3,875	4,000	7,875
Total	5,216	8,000	13,216

Figure 3 shows the visual representation of different classes of images.

2) *Data Preprocessing*: Pre-processing follows the steps:

- CLAHE and Laplacian filter: enhance image quality.
- SMOTE: Is used for balancing the dataset.

3) *Feature Extraction*: CNN-based feature extractors are used here, which are already pre-trained on the ImageNet dataset. DenseNet201, DenseNet201, EfficientNetB0, and ResNet152V2 are CNN model that takes input (224×224×3).

4) *Classifications*: A Custom Deep learning classifier helps to get the best result. In our classifier, we have many custom layers, which are connected. This model takes input as a CNN feature extractor model output, which is already pre-trained. This input helps the classifier to achieve decent results within a short time. The output of this binary classifier is a 1D array with one element. The algorithm of the custom classifier is shown in Algorithm 1.

Algorithm 1 Feature Extraction and Classification Using Pretrained CNN Model

Require: Pretrained DenseNet201 model

Ensure: A trainable binary classification model

Step 1: Load Feature Extractor

- 1: Load DenseNet201 with ImageNet weights
- 2: Set `include_top = False`, input shape = (224, 224, 3)
- 3: Freeze initial layers to retain pre-trained features
- 4: Unfreeze last 80 layers for fine-tuning

Step 2: Extract Feature Maps

- 5: $input_tensor \leftarrow cnn_model.input$
- 6: $feature_maps \leftarrow cnn_model(input_tensor)$
- 7: $pooled \leftarrow GlobalAveragePooling2D()(feature_maps)$

Step 3: Build Custom Classifier

- 8: $dense1 \leftarrow Dense(512)(pooled)$
- 9: $norm1 \leftarrow BatchNormalization()(dense1)$
- 10: $act1 \leftarrow Activation('relu')(norm1)$
- 11: $drop1 \leftarrow Dropout(0.25)(act1)$
- 12: $output \leftarrow Dense(1, activation = 'sigmoid')(drop1)$

Step 4: Final Model

- 13: $model \leftarrow Model(inputs = input_tensor, outputs = output)$
- 14: Compile model with:
 - Loss: `binary_crossentropy`
 - Optimizer: Adam (learning rate = 1×10^{-4})
 - Metric: accuracy

- 15: **return** model output -[0.0]
-

IV. RESULTS & DISCUSSION

This section provides the outcome of the experiment with certain evaluation metrics. Additionally, a set of necessary graphs is provided.

A. Best Result

The best result is obtained by DenseNet201 and ResNet152V2 as the feature extractor, combined with a custom deep learning-based classifier :

- 1) Accuracy: **99.68%** for dataset-1[2] and **98.75%** for dataset-2 [14]. It indicates how many predictions are correct.
- 2) Precision: **99.68%** for dataset-1[2] and **99.68%** for dataset-2 [14]. It indicates how many predicted as true positives are actual positives.
- 3) Recall: **99.68%** for dataset-1[2] and **99.68%** for dataset-2 [14]. It indicates how many actual positive models correctly identify
- 4) F1-Score: **99.68%** for dataset-1[2] and **99.68%** for dataset-2 [14]. It indicates that Precision and recall are both high, and the model is balanced
- 5) AUC: **99.68%** for dataset-1[2] and **99.68%** for dataset-2 [14]. In measuring AUC, for dataset-1, the DenseNet121 with the custom classifier provides the best result. It measures the model's ability to distinguish between classes.

TABLE II: Comparison of All Employed Feature Extractor & Classifier Combinations on Two Datasets

Dataset	Feature Extractor	Classifier	Acc (%)	Precision (%)			Recall (%)			F1 Score (%)			AUC (%)
				0	1	w_avg	0	1	w_avg	0	1	w_avg	
Dataset 2 (Balanced) [14]	DenseNet121	Custom	98.56	98.99	98.14	98.57	98.12	99.00	98.56	98.56	98.57	98.56	99.92
		XGB	96.44	96.50	96.38	96.44	96.37	96.50	96.44	96.44	96.44	96.44	99.46
		SVM	96.81	97.35	96.29	96.82	96.25	97.38	96.81	96.79	96.83	96.81	99.13
	DenseNet201	Custom	98.38	98.01	98.74	98.38	98.75	98.00	98.38	98.38	98.37	98.37	99.87
		XGB	95.50	95.96	95.05	95.50	95.00	96.00	95.50	95.48	95.52	95.50	99.38
		SVM	94.75	95.78	93.77	94.77	93.63	95.87	94.75	94.69	94.81	94.75	99.20
	EfficientNetB0	Custom	98.12	97.89	98.37	98.13	98.38	97.88	98.12	98.13	98.12	98.12	99.80
		XGB	90.44	92.62	88.47	90.54	87.88	93.00	90.44	90.19	90.68	90.43	97.30
		SVM	67.00	75.19	62.83	69.01	50.75	83.25	67.00	60.60	71.61	66.10	72.87
	ResNet152V2	Custom	98.75	98.63	98.87	98.75	98.88	98.62	98.75	98.75	98.75	98.75	99.87
		XGB	96.06	96.59	95.55	96.07	95.50	96.63	96.06	96.04	96.08	96.06	99.26
		SVM	96.19	96.25	96.13	96.19	96.13	96.25	96.19	96.19	96.19	96.19	99.27
Dataset 1 [2]	DenseNet121	Custom	99.03	98.74	99.61	99.04	99.61	98.45	99.03	99.04	99.03	99.03	99.96
		XGB	98.77	98.09	99.48	98.78	99.48	98.06	98.77	98.78	98.77	98.77	99.89
		SVM	99.10	98.35	99.87	99.11	99.87	98.32	99.10	99.10	99.09	99.10	99.94
	DenseNet201	Custom	99.68	99.74	99.61	99.68	99.61	99.74	99.68	99.68	99.68	99.68	100.00
		XGB	98.77	98.21	99.35	98.78	99.35	98.19	98.77	98.78	98.77	98.77	99.93
		SVM	98.45	97.71	99.21	98.46	99.23	97.68	98.45	98.46	98.44	98.45	99.87
	EfficientNetB0	Custom	56.77	57.08	57.08	57.08	56.77	56.77	56.77	56.31	56.31	56.31	57.35
		XGB	94.00	91.79	96.46	94.12	96.65	91.35	94.00	94.15	93.84	94.00	98.17
		SVM	84.19	79.51	90.64	85.08	92.13	76.26	84.19	85.36	82.83	84.09	88.64
	ResNet152V2	Custom	98.97	98.10	99.87	98.98	99.87	98.06	98.97	99.98	98.96	98.97	99.95
		XGB	98.71	98.09	99.35	98.72	99.35	98.06	98.71	98.72	98.70	98.71	99.90
		SVM	99.16	98.47	99.87	99.17	99.87	98.45	99.16	99.17	99.16	99.16	99.95

Note: Class label **0** = Healthy, **1** = Pneumonia. **w_avg**=Weighted Average

This result shows my model can accurately distinguish normal and pneumonia-affected chest X-ray images.

TABLE III: Optimized Hyper-parameters

Classifiers	Hyper-parameters
Custom DL-based Classifier	ep: 31, ilr: 1e-4, act: relu + sigmoid
SVM	'C': 10, 'G': 0.01, 'K': 'rbf'
XGBoost	n: 100, m_depth: 5, lr: 0.1, em: 'logloss'

B. Result Table

Performance Comparison Table As shown in Table II, the output of the experiment.

In Table III, show the Hyperparameters according to each of the classifiers.

C. Comment

DenseNet201 and ResNet152V2 are the best-performing feature extractors. A custom classifier helps to achieve optimal results.

D. Performance

Figure 4 illustrates a comparative evaluation of model performance using two datasets. Subfigures (a), (b), and (c) result from Dataset 1, while subfigures (d), (e), and (f) correspond to Dataset 2.

In subfigures (a) and (d), show the Confusion Matrix, which helps to visualize how correctly our model classifies. Subfigures (b) and (e) show a balance between precision

and recall. In subfigures (c) and (f) ROC Curve represents a classifier's performance across all classifications.

In table IV, the performance of our proposed model is compared with the related model from the previous study.

V. LIMITATION & FUTURE WORK

In this section, the shortcomings of the research are described. Additionally, the future plan is outlined.

A. Limitation

This proposed study achieved decent accuracy, but it has some limitations. This experiment meets limited computational resources, which restricts the exploration of a large dataset. This model was trained and tested on a publicly available dataset only. When it comes to working on a real-world dataset, a model can face difficulty in evaluation.

B. Future work

In the future, we will work with Five-fold cross-validation that ensures more robust performance evaluation. Integration of clinical metadata to enhance prediction accuracy. Deployment in real-time systems or mobile applications that enable fast and more accessible pneumonia detection.

VI. CONCLUSION

Pneumonia is a lung infection that significantly impacts children and the elderly. In this study, we have demonstrated the effectiveness of deep learning and machine learning techniques in detecting pneumonia from chest X-ray images. We used

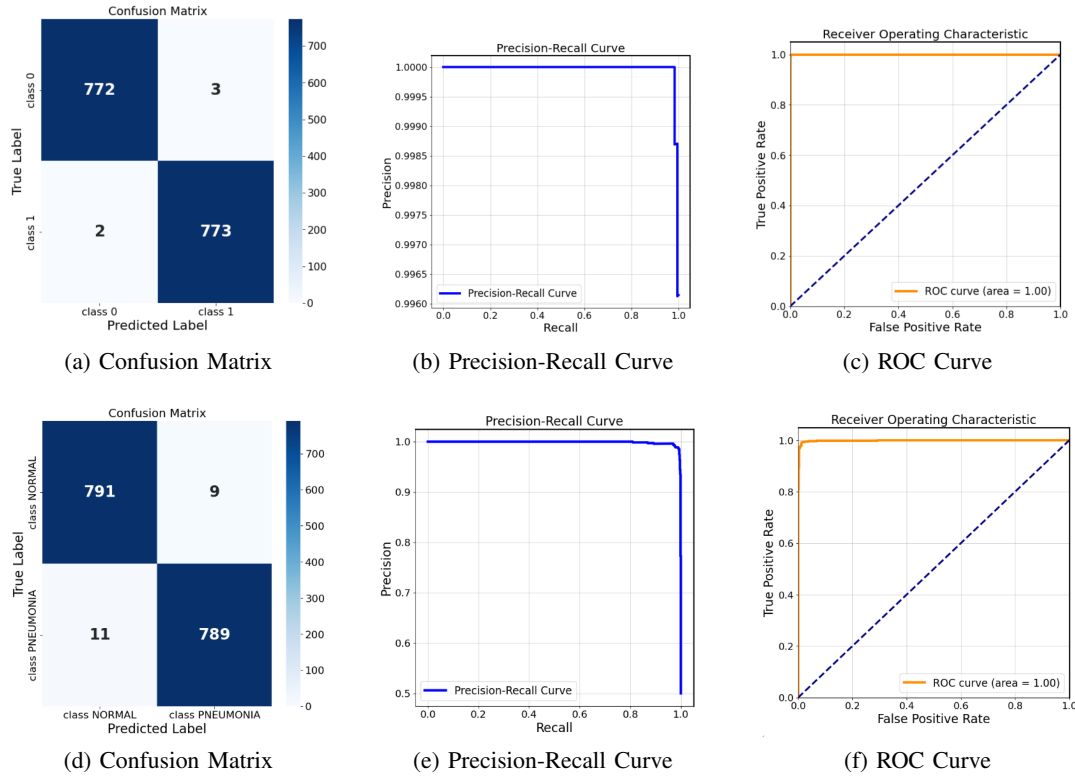


Fig. 4: Visual comparison of model performance. Subfigures (a), (b), and (c) are generated using **Dataset 1**, while subfigures (d), (e), and (f) are based on **Dataset 2**.

TABLE IV: Performance comparison of different models

Model	Acc (%)	Precision	Recall	F1-Score	AUC
DenseNet201+ Custom Classifier[Prop]	99.68	0.9974	0.9961	0.9968	1.0000
NN with VGG16 [3]	92.15	0.9428	0.9308	0.9370	0.9740
MobileNet [4]	94.23	—	—	—	—
eRBM [5]	99.30	0.9971	0.9989	2.9879	—
VGG16 [6]	87.50	0.8640	0.8810	0.8840	—
CNN [7]	96.80	—	—	0.7990	0.9300
Inception v3 [8]	97.20	—	—	—	—
Custom CNN [9]	97.00	0.9500	0.9820	0.9660	0.9800
ResNet 50V2 [10]	95.97	—	—	—	0.9800
Resnet34+maxvit_small [11]	94.87	—	1.00	—	—
Vision Transformer [12]	97.61	—	0.9500	—	—
Ensemble Learning(EL) [13]	93.91	0.9396	0.9299	0.9343	—

CNN models DenseNet121, DenseNet201, EfficientNetB0, and ResNet152V2 as feature extractors, which are combined with a custom deep learning classifier, XGBoost, and SVM as classifiers. In this study, we have used two publicly available datasets that contain healthy and pneumonia images. This paper combines deep learning and classical machine learning approaches to achieve fast detection of pneumonia.

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